Università degli
Studi di Milano-Bicocca

Introduction to Pytorch

Prof. Flavio Piccoli - Dr. Mirko Paolo Barbato

R&D process

- Analysis of the state of art
- Data collection / analysis
- **Architectural Design**
- Parameter Search
- Benchmarking
- Deployment

Google Scholar

pandas

scikit learn

PyTorch

PyTorch Lightning

RAY

tune

Streamlit

Flask

ONNX

Introduction to Pytorch

- machine learning framework based on the Torch library
- originally developed by Meta AI and now part of the Linux Foundation umbrella
- It is free and open-source software released under the modified BSD license
- Although the Python interface is more polished and the primary focus of development, PyTorch also has a C++ interface.[12]

PyTorch

Basic operations

- Convert a list in tensor

```
mat = torch.tensor([1,2,3,4,5])
```

- Get the shape of a tensor

```
mat.size()   or   mat.shape
```

```
torch.Size([5])
```

- Add a dimension to the tensor

```
mat = mat.unsqueeze(0)
```

```
torch.Size([1, 5])
```

- Remove singleton dimensions

```
mat = mat.squeeze()
```

Autograd

- Pytorch automatically performs derivatives for you
- Example: evaluate the derivative of $y = x^2$ in the point x=3

```
import torch
import torch.nn as nn

# define a number
x = torch.tensor([3.0])

# ask to compute gradients for that variable
x.requires_grad = True

# do an operation on the number
y = x**2

# perform backpropagation
y.backward()

# obtain gradients
print(x.grad)
```

It will print 6, because $y' = 2x$ and $x = 3$

Training procedure

The pseudocode corresponding to the full training procedure is:

1. initialize dataset (load the list of filenames and ground-tr.)
2. define predictive model
    a. initialize weights
3. define optimizer

Initial setup

4. for each epoch (or until convergence)
    a. for each batch in the dataset:
        i. compute forward propagation to get output
        ii. compute loss
        iii. compute gradients (back-propagation)
        iv. update weights of model

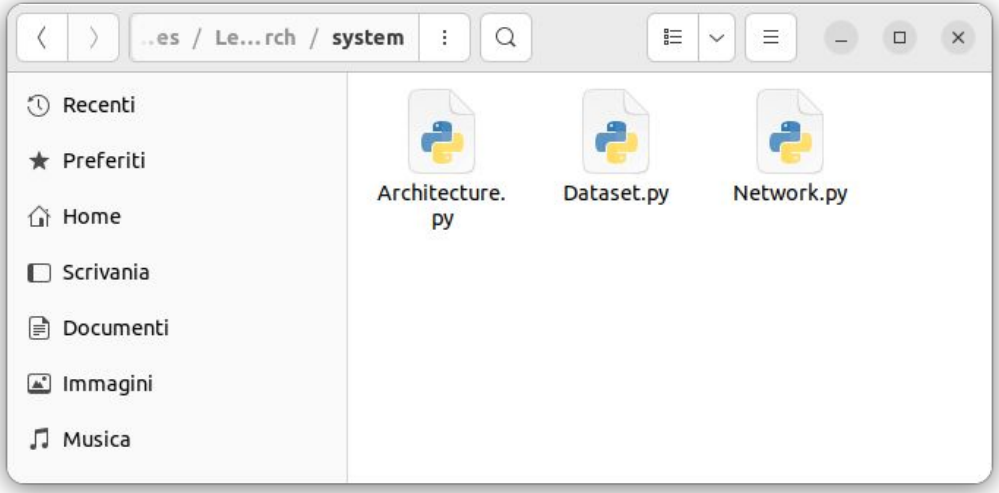
1-epoch training

    b. validate model
    c. if best model, save it

testing and checkpointing

File structure

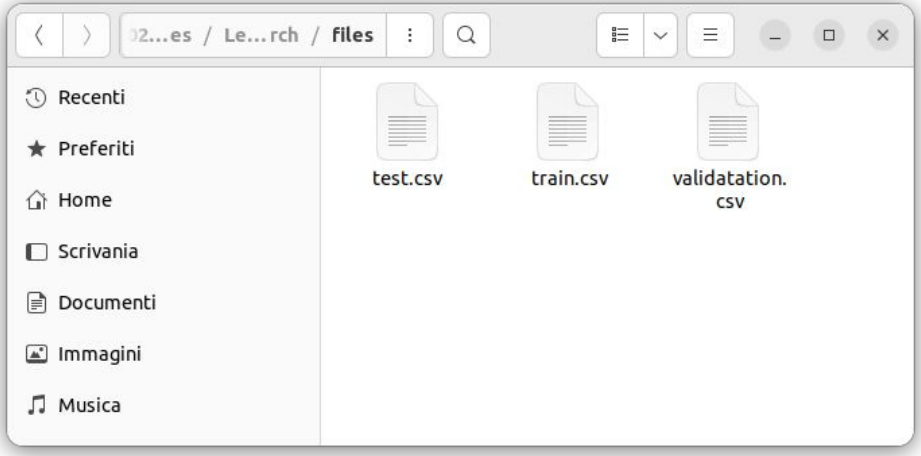
- It is a good practice to split in separate files:
  - data loading
  - network definition
  - architecture
- By doing so, each components can be tested separately without starting the whole system



Data loading for CSV files

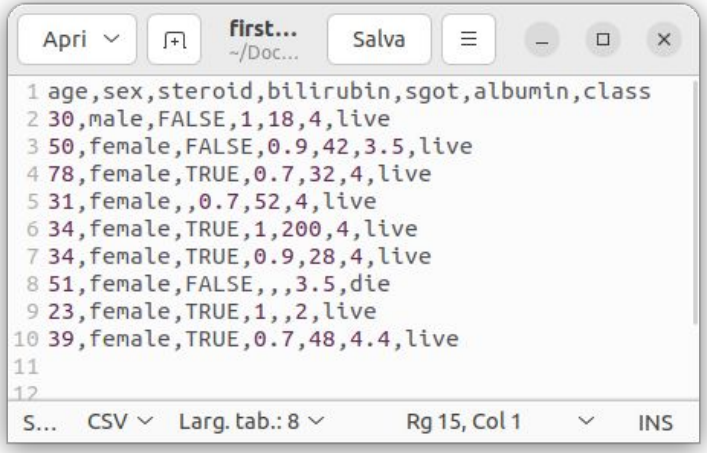
Data structure for CSV files

Data is splitted in three files, train, validation and test



Each CSV file is has:

- the first row containing the headers (opt.)
- the other rows are samples of the dataset



Data structure for CSV files

```
class Dataset(torch.utils.data.Dataset):
```

```
def __init__(self, csv):
    # here i will read my CSV file
    self.df = pd.read_csv(csv)
```

Initialization of the dataset

- read the csv file

```
def __len__(self):
    # here i will return the number of samples in the dataset
    return len(self.df)
```

Counting of the samples

- return the number of samples

```
def __getitem__(self, idx):
    # here i will load the file in position idx
    cur_sample = self.df.iloc[idx]
    # split in input / ground-truth
    cur_sample_x, cur_sample_y = split_input_output(cur_sample)
    # return values
    return cur_sample_x, cur_sample_y
```

Load and return item

- read csv line
- split input and ground-truth variables

Dataset usage

For a fast checking that everything is correct you can load a sample:

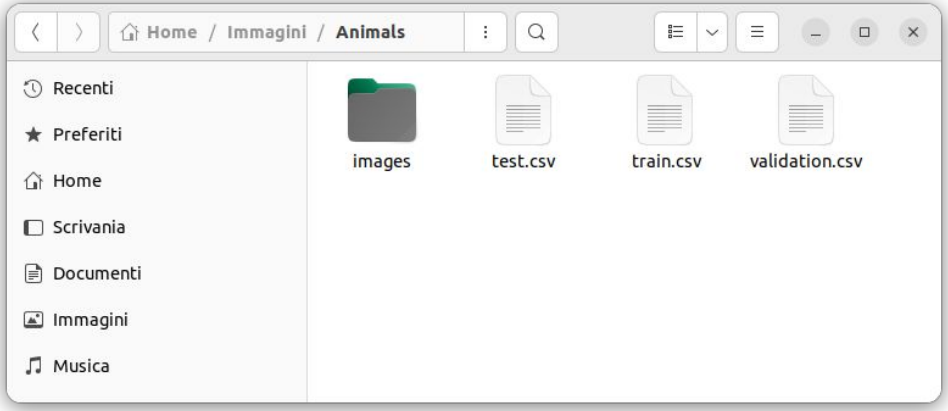
```
# init the dataset
ds = Dataset('/path/to/csv_file.csv')

# print number of samples
print(ds.__len__())

# load a sample to check that everything is fine
sample = ds.__getitem__(3)

# print it
print(sample)
```

Data loading for image datasets

Data structure for images

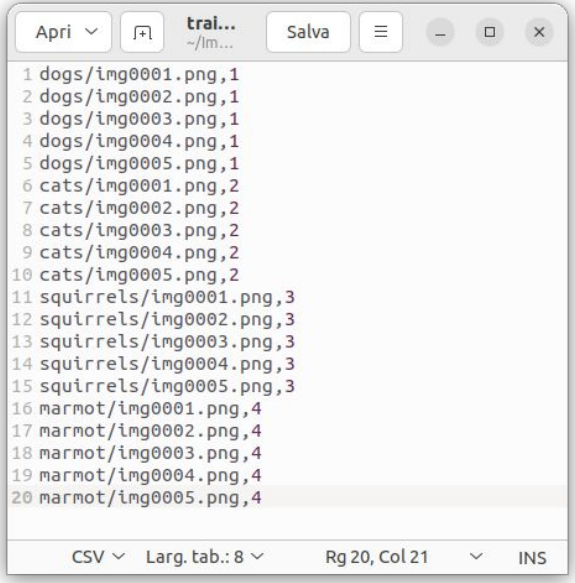
Generally, the dataset folder contains:

- a folder containing images
- three csv files containing the list of files
  - train
  - validation
  - test



Every line of the CSV contains:

- the path of the image
- the belonging class (ground truth)



Dataset for images

```
import os
import torch

class Dataset(torch.utils.data.Dataset):

    def __init__(self, csv):
        # here i will read my file list
        self.fns = ...


    def __len__(self):
        # here i will return the number of samples in the dataset
        return len(self.fns)


    def __getitem__(self, idx):
        # get current sample
        cur_sample = self.fns[idx]
        # split filename from class
        cus_fn, cur_class = cur_sample.split(',')
        # load the image
        loaded_file = load_file(cur_fn)
        # return image and ground-truth
        return loaded_file, cur_class
```

Dataset for images

```
import os
import torch

class Dataset(torch.utils.data.Dataset):
```

```
def __init__(self, csv):
    # here i will read my file list
    self.fns = ...
```

Initialization of the dataset

- read file list

```
def __len__(self):
    # here i will return the number of samples in the dataset
    return len(self.fns)
```

Counting of the samples

- return the number of samples

```
def __getitem__(self, idx):
    # get current sample
    cur_sample = self.fns[idx]
    # split filename from class
    cus_fn, cur_class = cur_sample.split(',')
    # load the image
    loaded_file = load_file(cur_fn)
    # return image and ground-truth
    return loaded_file, cur_class
```

Load and return item

- get string containing filename and gt
- separate filename from gt
- load image
- return image and ground-truth

A simple example

Create a dataloader with 2 inputs:

- a CSV file with filenames
- a directory containing images

The dataloader must:

- read the csv in __init__
- in __getitem__:
  - split filename from gt
  - load image

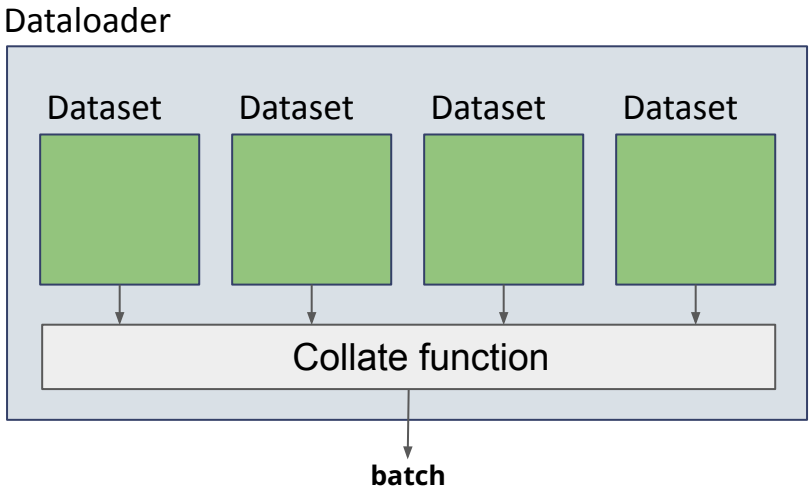
```python
import os
import torch
from torchvision import transforms
import numpy as np
from PIL import Image


class AnimalDataset(torch.utils.data.Dataset):

    def __init__(self, root_dir, csv_file):
        # save root directory
        self.root_dir = root_dir
        # read file
        f = open(csv_file, "r")
        txt = f.read()
        f.close()
        # get filenames
        self.fns = txt.split('\n')


    def __len__(self):
        return len(self.fns)


    def __getitem__(self, idx):
        # get current sample
        cur_sam = self.fns[idx]
        # split filename from class
        cur_fn, cur_gt =  cur_sam.split(',')
        # appen root path
        cur_fn = os.path.join(self.root_dir, cur_fn)
        # open image
        cur_img = Image.open(cur_fn)
        # convert to pytorch
        cur_img = transforms.ToTensor()(cur_img)
        # return
        return cur_img, cur_gt
```

Parallel data loading

Parallelize your dataset to load multiple instances at once

- create n copies of your dataset
- uses each copy to read a sample
- joins the loaded samples in collate function

```
# init dataset
dataset = AnimalDataset('images','train.csv')

# define dataloader
dataloader = torch.utils.data.DataLoader(
            dataset,
            batch_size=80,
            drop_last=False,
            shuffle=False,
            num_workers=8
        )

# use dataloader
for cur_images, cur_gts in dataLoader:
    # for each batch do something
    print(cur_images.shape)
```

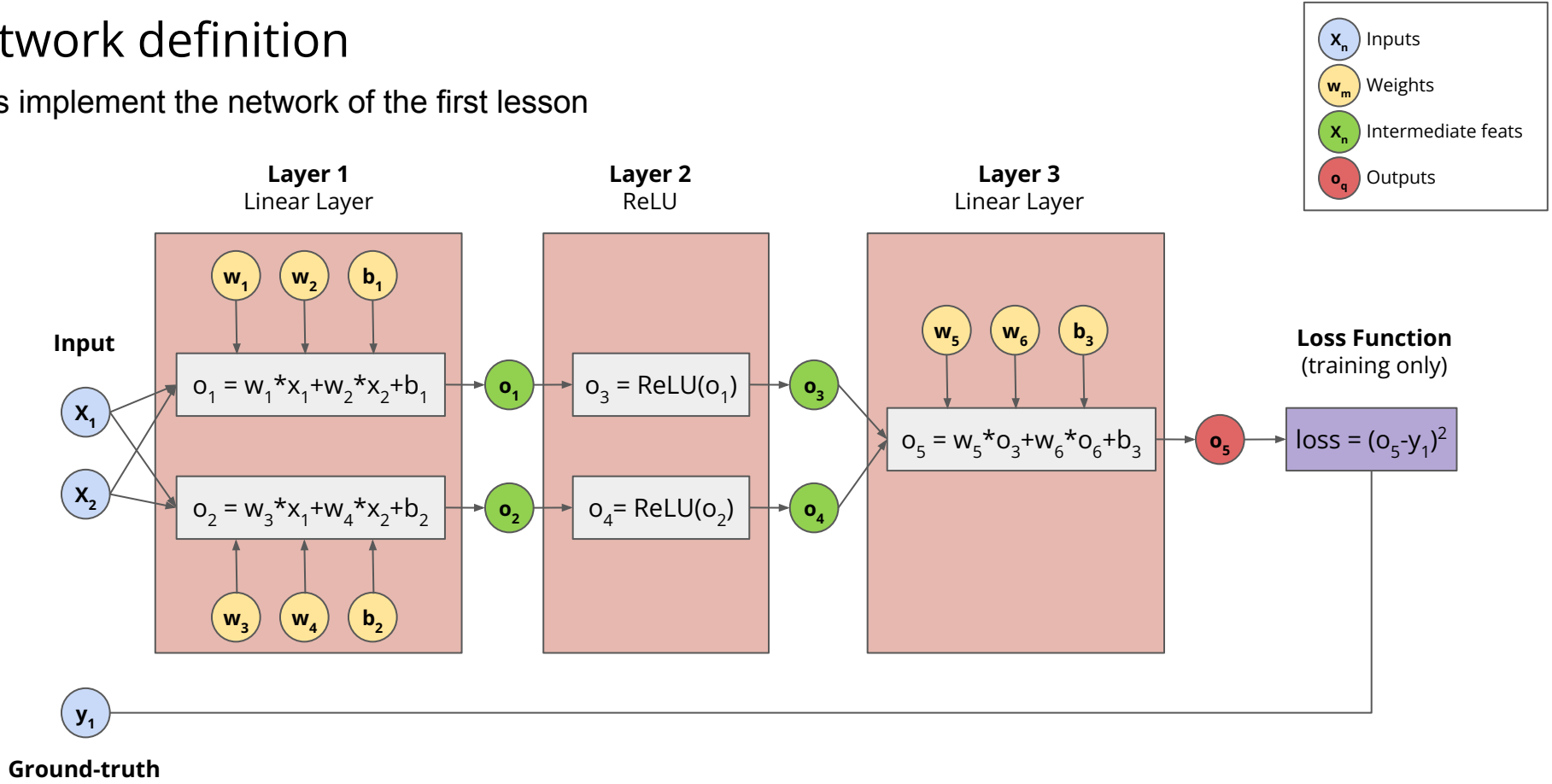


- defined with torch.utils.data.DataLoader
- you need to specify:
  - *batch_size*: dimension of the batch
  - *num_workers*: number of threads
  - *drop_last*: last batch must be dropped if uncomplete
  - *shuffle*: shuffle instances at each epoch

Network definition

Network definition

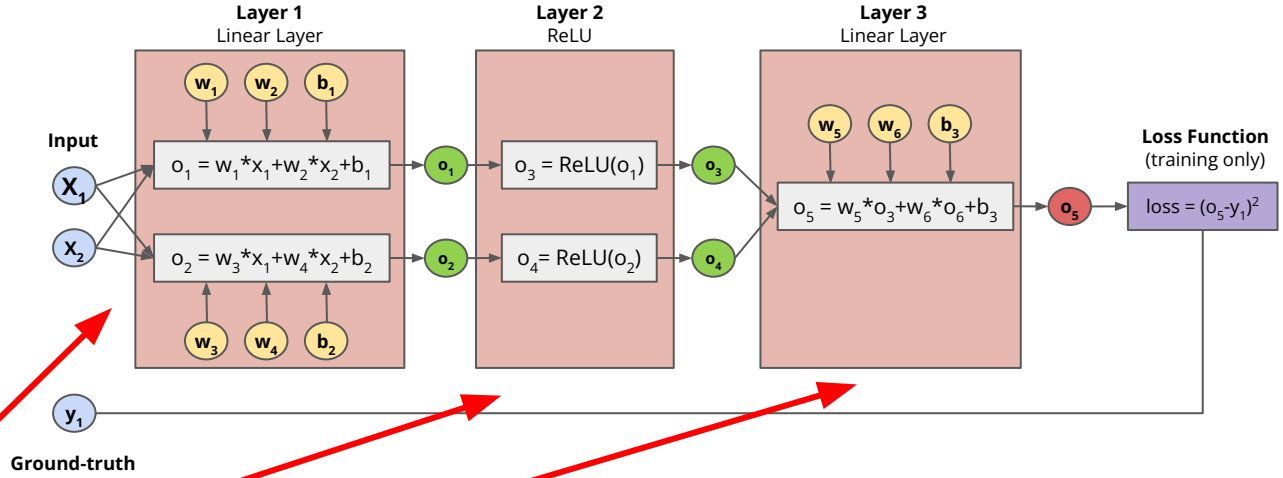
Let's implement the network of the first lesson



Network definition

Let's implement the network of the first lesson

```
import os
import torch
import torch.nn as nn


class Network(nn.Module):

    def __init__(self):
        # initialize super class
        super(Network, self).__init__()

        # create first layer
        self.layer1 = nn.Linear(2,2)
        # create second layer
        self.layer2 = nn.ReLU()
        # create third layer
        self.layer3 = nn.Linear(2,1)


    def forward(self, x):
        # apply layers in cascade
        x = self.layer1(x)
        x = self.layer2(x)
        x = self.layer3(x)
        # return output
        return x
```



The network definition is performed by extending the nn.Module class. Must have two methods:

- __init__
- forward

nn.Linear(**n_inp**, **n_out**)

Network definition

```
import os
import torch
import torch.nn as nn


class Network(nn.Module):

    def __init__(self):
        # initialize super class
        super(Network, self).__init__()

        # create first layer
        self.layer1 = nn.Linear(2,2)
        # create second layer
        self.layer2 = nn.ReLU()
        # create third layer
        self.layer3 = nn.Linear(2,1)

    def forward(self, x):
        # apply layers in cascade
        x = self.layer1(x)
        x = self.layer2(x)
        x = self.layer3(x)
        # return output
        return x
```

Initialization of the network

- define layers

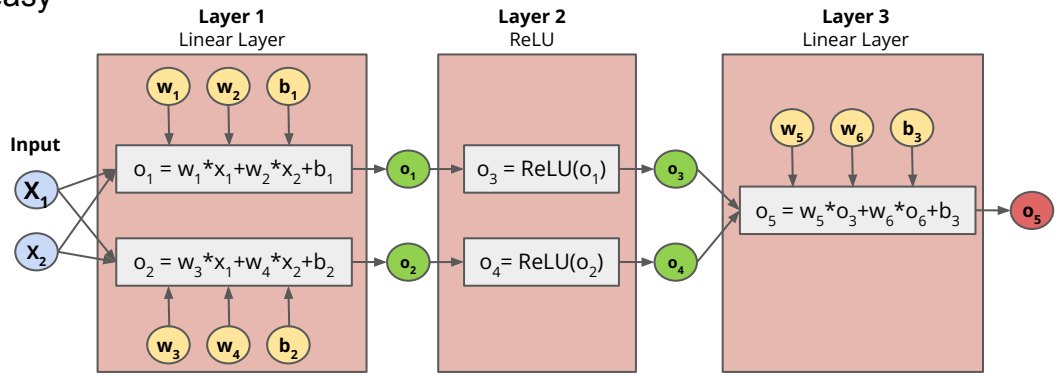
Use of the network

- define how to use the layers
- from input to output

Example of network usage

Instantiating and using a network in Pytorch is extremely easy

```
# initialize network
net = Network()

# create fake input
inp = torch.rand(20, 2)

# apply network
out = net(inp)

# print output size
print(out.size())
```



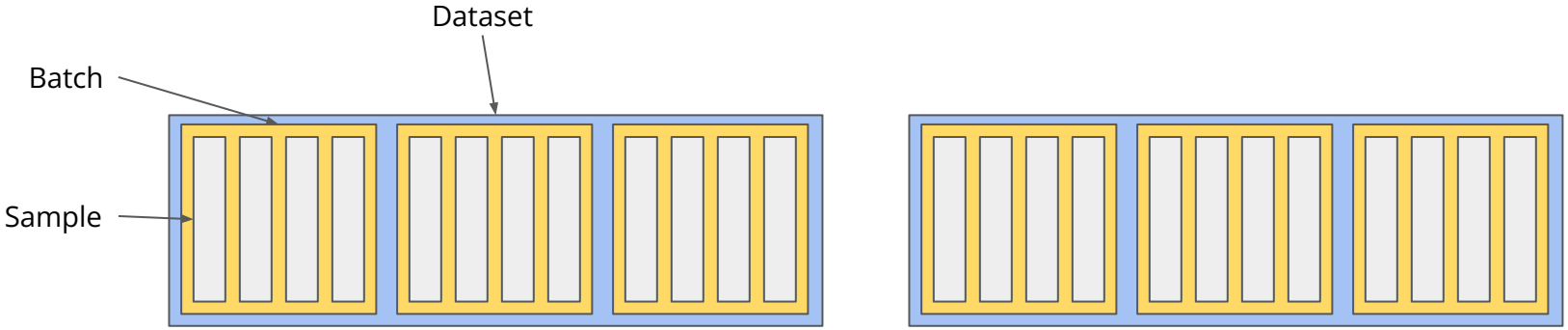

```
----------------------------------------------------------------
        Layer (type)               Output Shape         Param #
================================================================
            Linear-1                [-1, 20, 2]               6
              ReLU-2                [-1, 20, 2]               0
            Linear-3                [-1, 20, 1]               3
================================================================
Total params: 9
Trainable params: 9
Non-trainable params: 0
----------------------------------------------------------------
Input size (MB): 0.00
Forward/backward pass size (MB): 0.00
Params size (MB): 0.00
Estimated Total Size (MB): 0.00
----------------------------------------------------------------
```

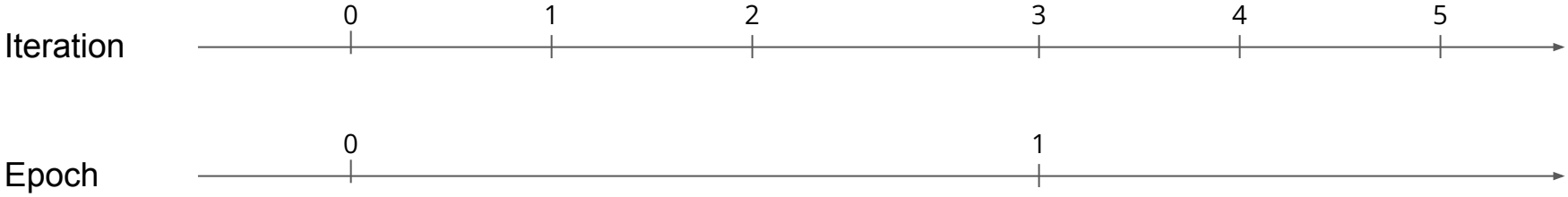
System definition

Terminology

Iteration = processing of one batch

Epoch = processing of the entire training set





Initial setup

```python
import os
import torch
from Network import Network
from Dataset import Dataset
```

```python
# initialize datasets
train_dataset = Dataset('./dataset/train.csv')
val_dataset = Dataset('./dataset/validation.csv')

# create training dataloader
train_dataloader = torch.utils.data.DataLoader(
    train_dataset,
    batch_size = 20,
    drop_last = True,
    shuffle = True,
    num_workers = 8
)

# create validation dataloader
val_dataloader = torch.utils.data.DataLoader(
    val_dataset,
    batch_size = 20,
    drop_last = False,
    shuffle = False,
    num_workers = 8
)
```

Definition of the datasets and dataloaders

```python
# create network
net = Network()

# define optimizer
optimizer = torch.optim.SGD(params=net.parameters(), lr=0.001)
```

Definition of the network and the optimizer

Training cycle

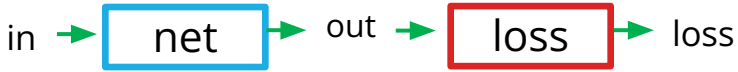
```
# for each epoch
for cur_epoch in range(1000):

    # for each batch
    for inp, gt in train_dataloader:

        # reset gradients
        optimizer.zero_grads()
```

1. Reset grads

```
        # forward
        out = net(inp)
```

2. Forward

```
        # compute loss
        loss = F.mse_loss(out, gt)
```



3. Loss

```
        # backward
        loss.backward()
```

4. Backward

```
        # update weights
        optimizer.step()
```

5. Update weights

Saving objects on disk (single object)

- With Pytorch is very easy to save stuff on disk
  - torch.save(tensor or dictionary, filename)
  - torch.load(filename)

```
# define a tensor
mat = torch.tensor([1,2,3,4,5])
# save it to disk
torch.save(mat, 'mat.pth')
# reload it
new_mat = torch.load('mat.pth')
```

Saving objects on disk (dictionary of objects)

- We can also save dictionaries with Pytorch
  - good for saving multiple tensors in a single file

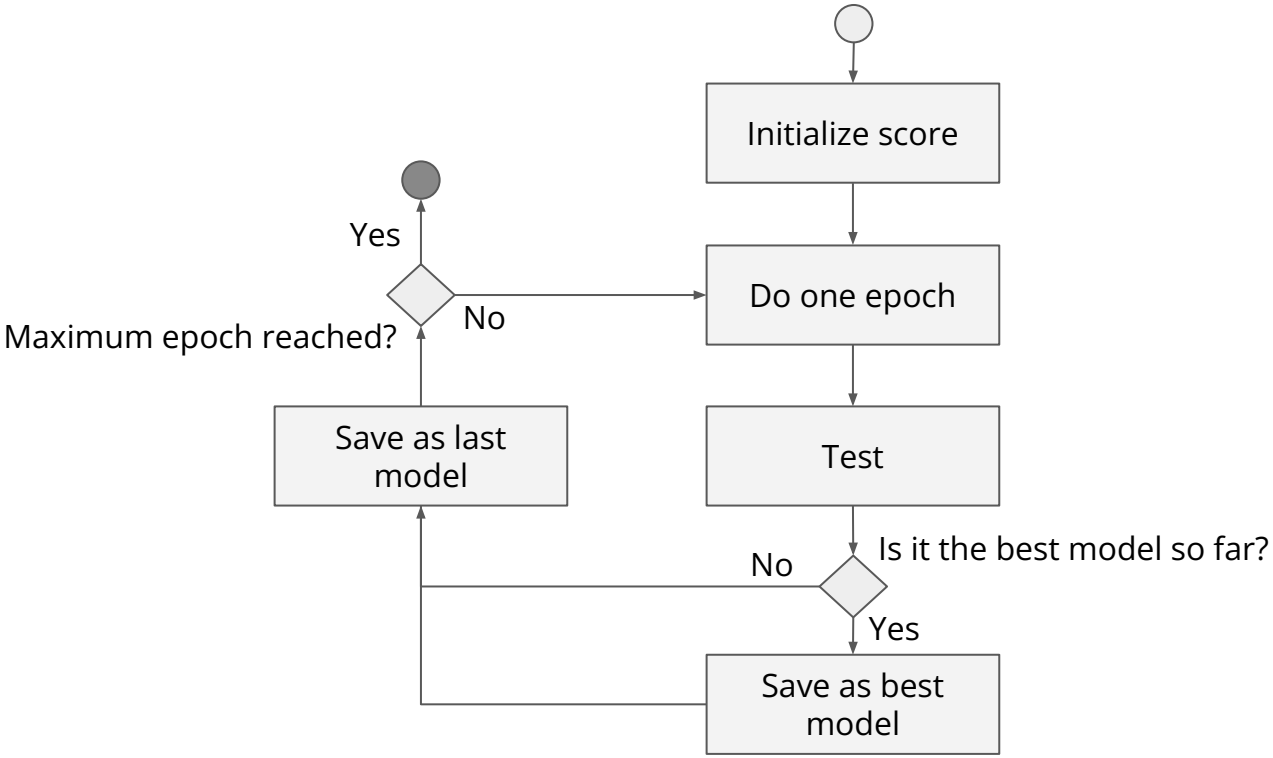
```
# define two tensors
mat1 = torch.tensor([1,2,3,4,5])
mat2 = torch.tensor([6,7,8,9,10])

# create dictionary
d = {
    'mat1': mat1,
    'mat2': mat2,
    'int1': 3,
    'string1': 'ciao',
}

# save it to disk
torch.save(d, 'dict.pth')

# reload it
new_d = torch.load('dict.pth')

# use a variable
new_mat1 = new_d['mat1']
```

Saving network and optimizer state on disk

- We will use the saving function to save the state of the network and the optimizer

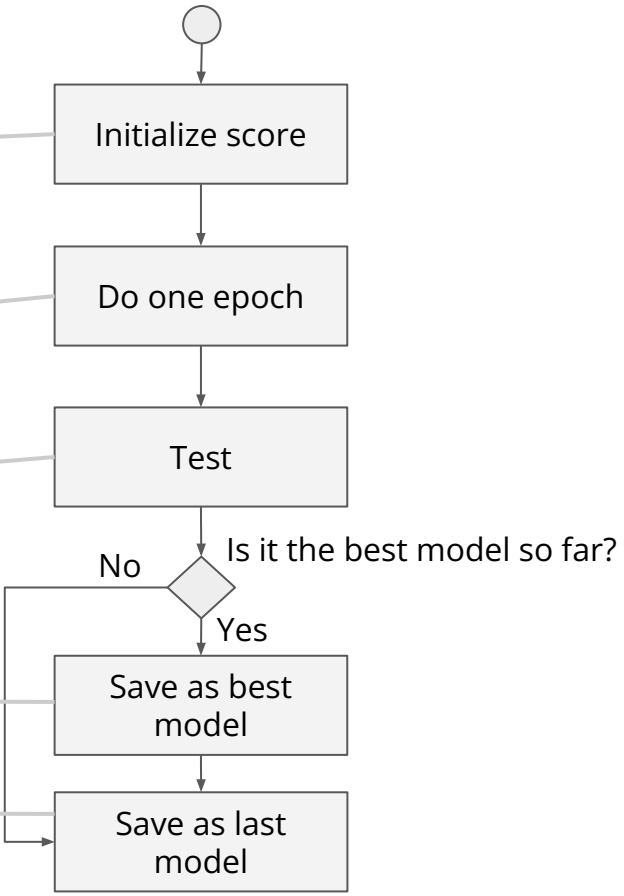
```
# initialize network
net = Network()

# initialize optimizer
opt = torch.optim.SGD(params=net.parameters(), lr=0.001)

# create dictionary containing states
d = {
    'net': net.state_dict(),
    'opt': opt.state_dict()
}

# save state
torch.save(d, 'last.pth')

# load state from disk
new_d = torch.load('last.pth')

# load parameters of network and optimizer
net.load_state_dict(new_d['net'])
opt.load_state_dict(new_d['opt'])
```

Checkpointing



Checkpointing (example)

```
# initialize best accuracy
best_accuracy = 0

# for each epoch
for cur_epoch in range(1000):
    # for each batch
    for inp, gt in train_dataloader:
        # … process batch and update weights

    # at the end of the epoch, test the model
    cur_accuracy = test_the_model(model, val_data_loader)

    # check if it is the best model
    if cur_accuracy > best_accuracy:
        # if yes, set new best accuracy and save best model
        best_accuracy = cur_accuracy
        save_model(model, best=True)

    # save last model
    save_model(model, best=False)
```



Moving tensors and networks in GPU

You can easily move tensors and networks on device with:

```
# specify device
device = 'cuda'

# move tensor to device
mat = mat.to(device)

# move network to device
net.to(device)
```

Usually, you:

- move the network to GPU after network definition
- move the batch in GPU at the beginning of each iteration

Quiz Time

Quiz time!

Go to the website:

PollEv.com/flaviopiccoli014

How do you convert a list into a tensor?

myTensor = torch .convert_to_tensor(myList)

myTensor = tensor(myList)

myTensor = torch.tensor(myList)

myTensor = myList.to_tensor()

Powered by Poll Everywhere

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

How do you get the shape of a tensor?

tensor.size()

tensor.shape()

tensor.shape

tensor.size

Powered by Poll Everywhere

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Which methods must implement a class that extends torch.utils.data.Dataset?

__getitem__

__len__ and __getitem__

__init__, __len__ and __getitem__

__init__, __len__ , __collate__ and __getitem__

Powered by Poll Everywhere

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Why do you need to convert a dataset into a dataloader?

To make it torch compatible

For fast testing

For parallel dataloading

For integrating the code in the pipeline

Powered by Poll Everywhere

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

What does the parameter num_workers of torch.utils.data.DataLoader specify?

The number of threads used for parellel loading

The number of items in a batch

The number of parameters in the network

The dimension of the batch

Powered by Poll Everywhere

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

What does the parameter drop_last of torch.utils.data.DataLoader mean?

Use last batch only in validation

Drop the last batch to keep it for testing

If the last batch is not big enough to cover the batch size, drop it

Put spurious data in the last batch and don't consider it during training

Powered by Poll Everywhere

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

How do you specify a linear layer of 20 neurons that takes 10 inputs? What will be the size of its output? (B=batch size)

nn.Linear(20, 10) and its output will be (20)

nn.Linear(10, 20) and its output will be (B, 20)

nn.Linear(20, 10) and its output will be (B, 20)

nn.Linear(10, 10) and its output will be (B, 10)

Powered by Poll Everywhere

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

How do you move a tensor in GPU?

myTensor = myTensor.to('cuda')

myTensor.to('cuda')

myTensor.to('gpu')

myTensor = myTensor.to('gpu')

Powered by Poll Everywhere

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

How do you move a neural network in GPU?

myNet = myNet.to('cuda')

myNet.to('cuda')

myNet.to('gpu')

myNet = myTensor.to('gpu')

Powered by Poll Everywhere

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

What is an epoch?

The training process

The processing of a batch

The processing of a sample

The processing of the entire training set

Powered by Poll Everywhere

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Regression exercises

Google Colab

- University's computers do not have hardware acceleration (GPU)
- We need to use Google Colab

https://colab.research.google.com/

- It's the same as using a Notebook, but it is possible to add a GPU for parallel processing
- To enable:
  - Edit > Notebook settings as the following: Click on "Notebook settings" and select "GPU"

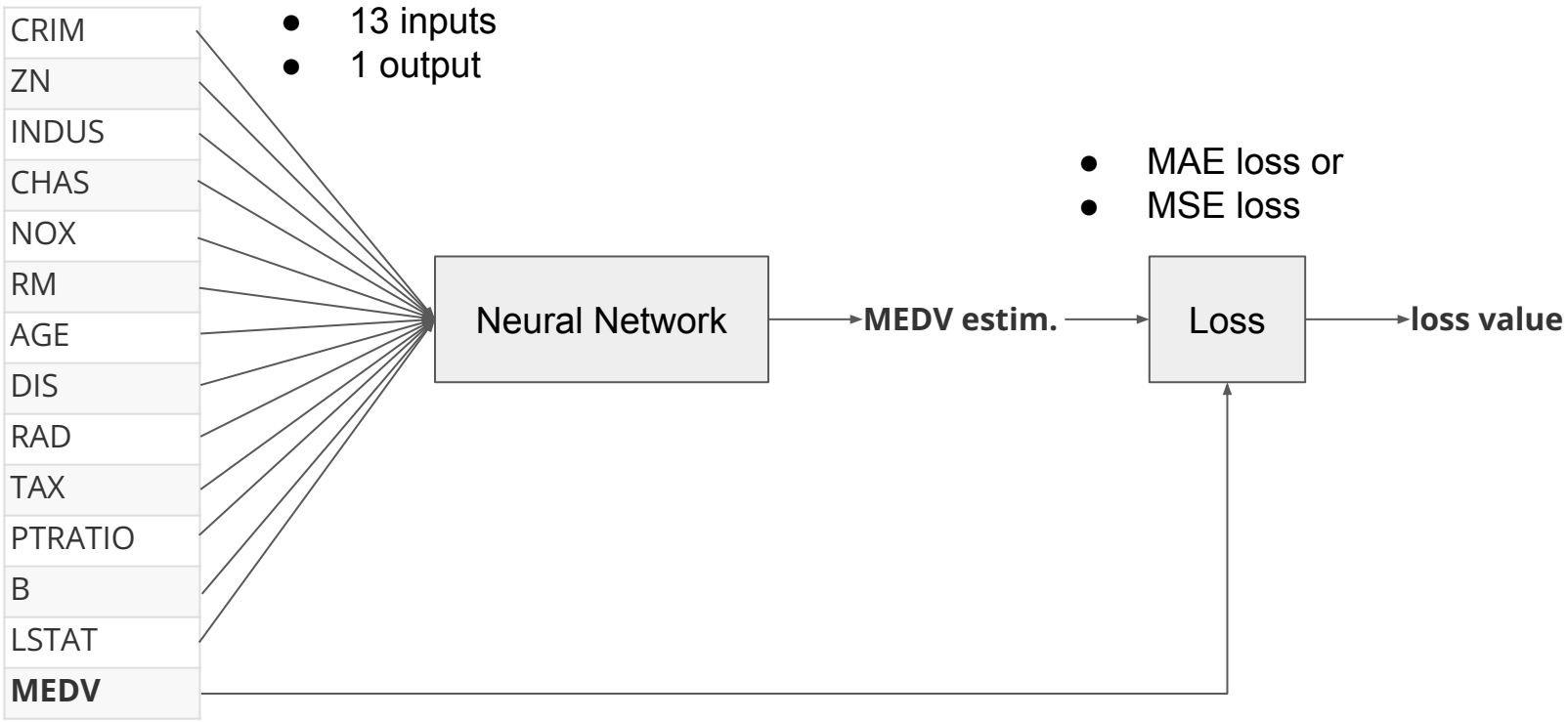
Exercise 1 - regression

Given the Boston Housing Dataset, predict the variable MDEV from the variables

Variable	Description
CRIM	per capita crime rate by town
ZN	proportion of residential land zoned for lots over 25,000 sq.ft.
INDUS	proportion of non-retail business acres per town
CHAS	Charles River dummy variable (= 1 if tract bounds river; 0 otherwise)
NOX	nitric oxides concentration (parts per 10 million)
RM	average number of rooms per dwelling
AGE	proportion of owner-occupied units built prior to 1940
DIS	weighted distances to five Boston employment centres
RAD	index of accessibility to radial highways
TAX	full-value property-tax rate per $10,000
PTRATIO	pupil-teacher ratio by town
B	1000(Bk - 0.63)^2 where Bk is the proportion of blacks by town
LSTAT	% lower status of the population
MEDV	Median value of owner-occupied homes in $1000's

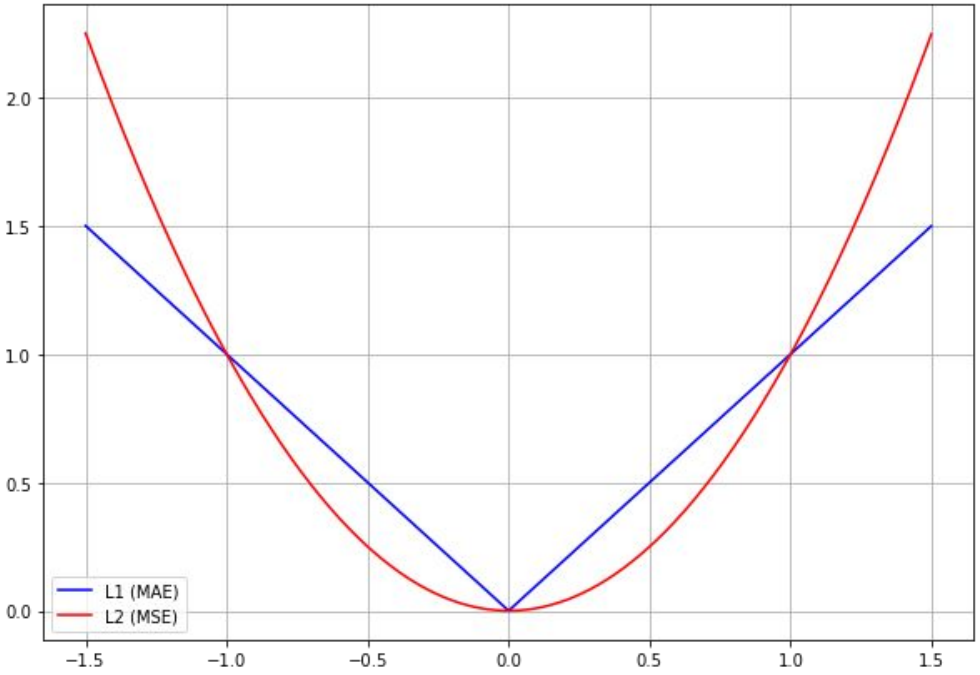
Exercise 1 - regression

System configuration



Exercise 1 - regression (together)

Losses for regression



$$\mathrm{MAE}(y, \hat{y}) = \frac{1}{N} \sum_{i=0}^{N-1} |y_i - \hat{y}_i|$$

$$\mathrm{MSE}(y, \hat{y}) = \frac{1}{N} \sum_{i=0}^{N-1} (y_i - \hat{y}_i)^2$$

$$y_i = ground\ truth$$

$$\hat{y}_i = estimated\ value$$

Exercise 1 - regression (together)

Operations:

1. create proper datasets and dataloaders
2. create a network
3. initialize optimizer
4. create training loop
5. use MAE (L1) both as training loss and metric for choosing the best model

Exercise 2 - regression

Predict Uber fare (var. **far_amount**). Replicate the same steps we performed in the previous exercise.
N.B.: data has already been pre-processed.

	fare_amount	pickup_longitude	pickup_latitude	dropoff_longitude	dropoff_latitude	passenger_count	year	Distance	month_1	month_2	...
0	7.5	-73.999817	40.738354	-73.999512	40.723217	1	2015	1681.11	0	0	...
1	7.7	-73.994355	40.728225	-73.994710	40.750325	1	2009	2454.36	0	0	...
2	12.9	-74.005043	40.740770	-73.962565	40.772647	1	2009	5039.60	0	0	...
3	5.3	-73.976124	40.790844	-73.965316	40.803349	3	2009	1661.44	0	0	...
4	16.0	-73.925023	40.744085	-73.973082	40.761247	5	2014	4483.73	0	0	...
...	...	...	...	...	...	...	...	...	...	...	...
199995	3.0	-73.987042	40.739367	-73.986525	40.740297	1	2012	112.13	0	0	...
199996	7.5	-73.984722	40.736837	-74.006672	40.739620	1	2014	1879.64	0	0	...
199997	30.9	-73.986017	40.756487	-73.858957	40.692588	2	2009	12867.92	0	0	...
199998	14.5	-73.997124	40.725452	-73.983215	40.695415	1	2015	3536.55	0	0	...
199999	14.1	-73.984395	40.720077	-73.985508	40.768793	1	2010	5410.68	0	0	...